

ANNEX – BIG CUBES AND THE MAP GREG AUSTIN

BIG CUBES 4x4 and 5x5
 Reduce, build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2n LAYER only – never widen it.

4x4 PLL edges
 4x4 OLL parity = 1 edge swapped – half the time
 4x4 PLL parity = 1 edge swap flipped – half the time

5x5 5x2 parity = 1 edge swap flipped – half the time **one move differs**. 5x5 has no PLL parity
 5x5 U2 x Rw U2 Rw U2 **3Rw** U2 Lw U2 Rw U2 Rw U2 Rw U2 Rw U2

NOTATION
 Each letter turns that face clockwise.
 U = from outside the cube – so from your seat. D and B look counter-clockwise = counter-clockwise. 2 = half turn.
 M = middle slice, turning the w L layer.
 x, y, z = the whole cube, turning the way R, U and F turn. Lowercase r = face plus the layer behind it, turning together.

WORDS
 algorithm = a fixed sequence of turns - trigger = a short algorithm who hands you as one unit - sexy move = the R or U or F - trigger - algorithm = the F R F R trigger. **edge pair** = two edges glued into one, as a 4x4 or 5x5

■ RURU' ■ fixed sequence ■ RURU' ■ MR'RF' edgeflapper gap = change grip

[illegible]

ANNEX – BIG CUBES AND THE MAP

BIG CUBES

4x4 and 5x5
 Reduce: build the 6 centers – pair the 12 edges –> solve it as a 3x3. Rw = 2 layers wide; 3Rw = 3 layers; 2R = 2nd LAYER ONLY – never hinder

4x4 PLL parity

1 edge pair flipped - half the time

Rw U2 x Rw U2 Rw U2 Rw' U2 Rw' U2 Rw' U2 Rw' U2 slice out, run it, slice back – both cubes

2R2 U2 2R2 Uw2 R2 R2 Uw2 2 edge pairs swapped - half the time

4x4 OLL parity

1 edge pair flipped - half the time

Rw U2 x Rw U2 Rw U2 Rw' U2 Rw' U2 Rw' U2 Rw' U2 slice out, run it, slice back – both cubes

Rw U2 x Rw U2 Rw U2 3Rw U2 1 edge pair flipped - half the time - **one move** differs; 5x5 has no PLL parity

NOTATION

Each letter turns that face clockwise, seen from outside the cube – so from your seat L and B look counter-clockwise = counter-clockwise. z = half turn.

U = middle slice, turning the way L turns. x: y z = the whole cube, turning the way R, U and F turn. Lowercase f = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT

use it until you know Sune

Twist a yellow corner four turns at a time, turning ONLY the top face between corners.

yellow faces RIGHT **R'D'R D2**

yellow faces FRONT **D'R'D x2**

ANNEX – BIG CUBES AND THE MAP

BIG CUBES 4x4 and 5x5
Reduce: build the 6 centers → pair the edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY – never widen it.

last 2 edges

Rw' U' R' U	R' F R F'	Rw	slide out, run it, slice back – both cubes
2R2 U2	2R2 Uw2	2R2 Uw2	2 edge pairs swapped – half the time

4x4 PLL parity

Rw U2 x Rw U2 Rw U2 Rw' U2	U2 Lw U2 Rw' U2	Rw U2 Rw' U2 Rw'
edge difference 3 slices = PLL parity		
Rw U2 x Rw U2 Rw U2 3Rw' U2	Lw U2 Rw' U2	Rw U2 Rw' U2 Rw'

Fix parity during the last layer, before OLL and PLL – both parity algorithms move corners.

NOTATION

Each letter turns that face clockwise, seen from outside the cube – so from your seat L and B look counter-clockwise. z = outer-clockwise. z' = half turn.
M = middle slice, turning the way L turns x y z = the whole cube, turning the R, U and F turn. Lowercase f f' = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT use it until you know Sune
Twist a yellow corner four turns at a time, turning ONLY the top face between corners.
yellow faces RIGHT **R'D'R D x2**
yellow faces FRONT **R'D'R x2**

WORDS
algorithms = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit - sexy move = the R U R' U' trigger - sledghehammer = the R' R F F' trigger - parity = a case a 3c3 can't contain - gap = two edges glued into one, on a 4x4 or 5x5

■ RURU' sexy move ■ R'RFRL Sune ■ RFRF' sledghehammer - gap - change grip